

Daniel DeLayo

Sixth-Year Computer Science Ph.D. Candidate

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Education

- 2021 – Present 📖 **Ph.D. Candidate, Computer Science**, Stony Brook University.
Research Interests: *Parallel Algorithms, Cache and Memory Management, Theory & Practice*
Advisor: *Michael A. Bender*
- 2017 – 2021 📖 **B.S. Computer Science**, Stony Brook University.
Honors: *Summa Cum Laude, Honors College, Honors Computer Science*.
GPA: 3.95/4.0

Employment History

- 2019 – Present 📖 **Research Assistant**, Stony Brook University, *Computer Science*.
- 2021 – 2024 📖 **Intern**, Sandia National Labs, *Cyber Security Algorithms Research*.
- 2018 – 2018 📖 **Research Assistant**, Stony Brook University, *Physics*.

Research Summary

My research focuses on how memory movement and contention limit performance, and on algorithms that overcome those limits. I develop parallel algorithms and space-efficient data structures that combine provable guarantees with high-performance implementations.

In caching, I study miss-ratio curves, caches with limited associativity, and caches with changing capacity. My algorithm Increment-and-Freeze computes exact miss-ratio curves in a single pass, up to $10\times$ faster than the previous best parallel algorithm. A sampling-based version is being integrated into WiredTiger, MongoDB's storage engine.

Beyond caching, I work on problems whose performance is dominated by memory movement: determinacy race detection in parallel programs, linear sketching for massive, dense graphs, and compact filters.

Research Publications

Journal Articles

- 1 D. Tench, E. West, V. Zhang, M. A. Bender, A. Chowdhury, **D. DeLayo**, J. A. Dellas, M. Farach-Colton, T. Seip, and K. Zhang. "GraphZeppelin: How to Find Connected Components (Even When Graphs Are Dense, Dynamic, and Massive)". In: *ACM Transactions on Database Systems* 49.3 (Sept. 2024), pp. 1–31. 📄 DOI: 10.1145/3643846.
- 2 J. Vorobyeva, **D. DeLayo**, M. A. Bender, M. Farach-Colton, P. Pandey, C. A. Phillips, S. Singh, E. D. Thomas, and T. M. Kroege. "Using Advanced Data Structures to Enable Responsive Security Monitoring". In: *Cluster Computing* 25.4 (Aug. 2022), pp. 2893–2914. 📄 DOI: 10.1007/s10586-021-03463-5.

Conference Proceedings

- 3 M. A. Bender, A. Conway, **D. DeLayo**, M. Farach-Colton, J. Han, L. He, R. Johnson, S. Kannan, W. Kuszmaul, D. Porter, and E. West. "Don't Melt Your Cache: Low-Associativity with Heat-Sink". In: *Proceedings of the 37th ACM Symposium on Parallelism in Algorithms and Architectures*. **Distinguished Paper Award**. ACM, July 2025, pp. 555–565. 📄 DOI: 10.1145/3694906.3743303.

- 4 J. L. Jones, A. T. Wilson, R. Gooding, **D. DeLayo**, K. Dalbey, J. Whetzel, and N. Sharan. “Behavioral Segmentation and Clustering of Geospatial Trajectories”. In: *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, Nov. 2025, pp. 4375–4381.  DOI: 10.1109/ITSC60802.2025.11423696.
- 5 D. Tench, E. T. West, K. Zhang, M. A. Bender, **D. DeLayo**, M. Farach-Colton, G. Gill, T. Seip, and V. Zhang. “Exploring the Landscape of Distributed Graph Sketching”. In: *2025 Proceedings of the Symposium on Algorithm Engineering and Experiments (ALENEX)*. Society for Industrial and Applied Mathematics, Jan. 2025, pp. 133–146.  DOI: 10.1137/1.9781611978339.11.
- 6 M. A. Bender, **D. DeLayo**, B. C. Kuszmaul, W. Kuszmaul, and E. West. “Increment-and-Freeze: Every Cache, Everywhere, All of the Time”. In: *Proceedings of the 35th ACM Symposium on Parallelism in Algorithms and Architectures*. ACM, June 2023, pp. 129–139.  DOI: 10.1145/3558481.3591085.
- 7 **D. DeLayo**, K. Zhang, K. Agrawal, M. A. Bender, J. W. Berry, R. Das, B. Moseley, and C. A. Phillips. “Automatic HBM Management: Models and Algorithms”. In: *Proceedings of the 34th ACM Symposium on Parallelism in Algorithms and Architectures*. ACM, July 2022, pp. 147–159.  DOI: 10.1145/3490148.3538570.

Preprints

- 8 Q. De Man, Q. Jafri, **D. DeLayo**, E. T. West, M. A. Bender, and D. Tench. *Fast and Compact Sketch-Based Dynamic Connectivity*. 2025.  DOI: 10.48550/ARXIV.2509.14433.

Software

- 2023 – Present  **Increment-and-Freeze**, fast parallel LRU hit-rate curve computation; being integrated into WiredTiger and OpenCilk.
github.com/etwest/Increment-and-Freeze
- 2025  **CUPCaKE**, parallel sketch-based dynamic connectivity for massive, dense graphs.
github.com/etwest/DynamicQueriesCC
- 2024  **Landscape**, distributed linear sketching for graph connectivity and k -edge connectivity.
github.com/GraphStreamingProject/Landscape
- 2022  **GraphZeppelin**, graph streaming library for connected components via linear sketching.
github.com/GraphStreamingProject/GraphZeppelin
-  **GutterTree (pipeline hypertree)**, parallel buffer-tree variant for batching graph-stream updates.
github.com/GraphStreamingProject/GutterTree

Awards

- 2026  **MongoDB PhD Fellowship**.
 **HALG Invited Talk**, Don’t Melt Your Cache: Low-Associativity with Heat-Sink.
- 2025  **SPAA Distinguished Paper Award**, Don’t Melt Your Cache: Low-Associativity with Heat-Sink.
- 2024 – 2025  **NSF Student Travel Grant**, Symposium on Parallelism in Algorithms and Architectures (SPAA).
- 2023  **CLSAC Invited Student Poster**, Chesapeake Large-Scale Analytics Conference.
- 2022  **GAANN Fellowship**, Stony Brook University.

Talks and Presentations

- 2026  **HALG (Invited)**, Don't Melt Your Cache: Low-Associativity with Heat-Sink.
-  **MAPSP**, Don't Melt Your Cache: Low-Associativity with Heat-Sink.
-  **MongoDB (PhD Fellowship)**, Increment-and-Freeze: LRU Miss-Ratio Curves for Every Cache, Everywhere, All of the Time.
- 2025  **SPAA**, Don't Melt Your Cache: Low-Associativity with Heat-Sink.
-  **NYU Theory Seminar**, Increment-and-Freeze: Every Cache, Everywhere, All of the Time.
-  **PhD Dissertation Proposal**, Practical Sequential and Parallel Paging via LRU Miss Ratio Curves.
- 2024  **PhD Prelim**, Increment-and-Freeze: Every Cache, Everywhere, All of the Time.
- 2023  **CLSAC**, Stateful Streaming with External Memory On Workstations.
- 2022  **SPAA**, Automatic HBM Management: Models and Algorithms.
-  **MAPSP**, Automatic HBM Management: Models and Algorithms.

Teaching Experience

- 2025  **Distributed Systems**, Teaching Assistant.
-  **Honors Algorithms**, Probability Lecture.
- 2023  **Graduate System Security**, Teaching Assistant.
- 2021  **Honors Algorithms**, Teaching Assistant.
- 2018  **Intro to Computer Science**, Teaching Assistant.

Professional Service

- 2026  **Shadow Program Committee**, SPAA.
-  **Artifact Evaluation (Reproducibility)**, ALENEX.
- 2025  **Shadow Program Committee**, SPAA.
-  **Artifact Evaluation (Reproducibility)**, ALENEX.
- 2024  **Artifact Evaluation (Reproducibility)**, ALENEX.
- 2023  **Peer Review**, ESA.
- 2022  **Peer Review**, IPDPS.